Diophantine Equations, SFFT, Bezout & Floor Functions Practice Exercises

1 How many ordered pairs of POSITIVE integers (x, y) satisfy the bilinear Diophantine equation $xy + 1x + 2y = 10$ (corresponding to positive lattice points on the plotted hyperbola)?

a. 6

b. 5

c. 3

d. 2

2 How many ordered pairs of POSITIVE integers (x, y) satisfy the bilinear Diophantine equation $xy + 2x + 3y = 18$ (corresponding to positive lattice points on the plotted hyperbola)?

a. 8

b. 3

c. 5

d. 2

3 How many ordered pairs of POSITIVE integers (x, y) satisfy the bilinear Diophantine equation $xy + 1x + 3y = 17$ (corresponding to positive lattice points on the plotted hyperbola)?

a. 3

b. 6

c. 5

d. 2

4 How many ordered pairs of POSITIVE integers (x, y) satisfy the bilinear Diophantine equation $xy + 2x + 5y = 26$ (corresponding to positive lattice points on the plotted hyperbola)?

a. 9

b. 5

c. 2

d. 3

5 How many ordered pairs of POSITIVE integers (x, y) satisfy the unit-fraction equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$?

a. 4

b. 5

c. 9

d. 8

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6 How many ordered pairs of POSITIVE integers (x, y) satisfy the unit-fraction equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{8}$?

a. 4

b. 7

c. 6

d. 8

7 How many ordered pairs of POSITIVE integers (x, y) satisfy the unit-fraction equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{9}$?

a. 5

b. 3

c. 4

d. 6

8 How many ordered pairs of POSITIVE integers (x, y) satisfy the unit-fraction equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{10}$?

a. 4

b. 5

c. 8

d. 9

9 Find the LARGEST positive integer n such that $n + 2$ divides $n^2 + 3n + 15$.

a. 13

b. 4

c. 11

d. 12

10 Find the LARGEST positive integer n such that $n + 3$ divides $n^2 + 3n + 17$.

a. 17

b. 14

c. 5

d. 15

11 Find the LARGEST positive integer n such that $n + 4$ divides $n^2 + 3n + 19$.

a. 19

b. 23

c. 7

d. 20

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12 Find the LARGEST positive integer n such that $n + 5$ divides $n^2 + 3n + 21$.

a. 31

b. 10

c. 27

d. 26

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